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Information processing apparatus, control method therefor, and storage medium

Abstract

An information processing apparatus includes at least one processor, and a memory that stores a program which, when executed by the at least one processor, causes the at least one processor to function as an acceptance unit configured to accept from a user a target impression of a poster image to be created, and a creation unit configured to create the poster image based on the target impression.

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Background/Summary

BACKGROUND OF THE DISCLOSURE

Field of the Disclosure

(1) The present disclosure relates to an information processing apparatus, a control method therefor, and a storage medium.

Description of the Related Art

(2) Traditionally, a method has been proposed to create a poster by preparing a template including information about shapes and positions of images, texts, and graphics that make up the poster, and automatically arranging the images, texts, and graphics according to the template.

(3) Japanese Patent No. 6537419 proposes generating a poster by selecting templates in ascending order of difference between the impression evaluation value of the template and the impression evaluation value of the image.

(4) In a technique disclosed in Japanese Patent No. 6537419, a template having a small difference between an impression evaluation value of the template and an impression evaluation value of an image is selected, but no consideration is given to creating a poster so as to express an impression intended by a user.

SUMMARY OF THE DISCLOSURE

(5) The present disclosure provides a technique for appropriately creating a poster expressing an impression intended by a user.

(6) In view of the above, the present disclosure provides an information processing apparatus including at least one processor, and a memory that stores a program which, when executed by the at least one processor, causes the at least one processor to function as an acceptance unit configured to accept from a user a target impression of a poster image to be created, and a creation unit configured to create the poster image based on the target impression.

(7) Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram illustrating a hardware configuration of a poster creation apparatus.

(2) FIG. 2 is a software block diagram of a poster creation application.

(3) FIG. 3A is a diagram illustrating an example of a skeleton, and FIG. 3B is a diagram illustrating an example of metadata.

(4) FIG. 4 is a diagram illustrating color scheme patterns.

(5) FIG. 5 is a diagram illustrating a display screen provided by a poster creation application.

(6) FIG. 6 is a diagram illustrating a display screen provided by a poster creation application.

(7) FIG. 7 is a flowchart illustrating a process of quantifying a poster impression in a poster creation process.

(8) FIG. 8 is a diagram illustrating a subjective evaluation of a poster.

(9) FIGS. 9A and 9B are flowcharts illustrating a poster creation process.

(10) FIGS. 10A to 10C are diagrams illustrating a skeleton selection method.

(11) FIG. 11A is a diagram illustrating color scheme patterns, and FIG. 11B is a diagram illustrating a font selection method.

(12) FIG. 12 is a software block diagram illustrating details of a layout unit.

(13) FIG. 13 is a flowchart illustrating a layout process.

(14) FIGS. 14A to 14C are diagrams illustrating inputs to a layout unit.

(15) FIGS. 15A to 15C are diagrams for illustrating an operation of a layout unit.

(16) FIGS. 16A to 16D are diagrams for illustrating examples of setting a target impression by a target impression specification unit.

- (17) FIG. 17 is a software block diagram of a poster creation application.
- (18) FIG. 18 is a flowchart illustrating a poster creation process.
- (19) FIGS. 19A to 19D are diagrams for illustrating a process by a combination generation unit.
- (20) FIGS. 20A and 20B are diagrams for illustrating a process by a combination generation unit.
- (21) FIG. 21 is a software block diagram of a poster creation application.
- (22) FIG. 22 is a flowchart illustrating a poster creation process.

DESCRIPTION OF THE EMBODIMENTS

(23) Embodiments of the present disclosure are described in detail below with reference to the drawings. The following embodiments are not intended to limit the disclosure, which is defined in the claims, and not all of the combinations of features described in the embodiments are essential to the disclosure. The same reference numbers are used for the same constituent elements, and duplicated descriptions thereof are omitted. The same reference numbers are used for the same constituent elements, and duplicated descriptions thereof are omitted.

First Embodiment

(24) A first embodiment discloses, by way of example, a method of automatically creating a poster by operating an application for creating a poster in a poster creation apparatus. In the following description, unless otherwise specified, an “image” refers to a still image or a frame image extracted from a moving image.

(25) FIG. 1 is a block diagram illustrating a hardware configuration of a poster creation apparatus **100**. The poster creation apparatus **100** may be realized by an information processing apparatus such as a personal computer (hereinafter referred to as “PC”), a smartphone, or the like. In the following description of the present embodiment, it is assumed that the poster creation apparatus is a PC. The poster creation apparatus **100** includes a CPU **101**, a ROM **102**, a RAM **103**, an HDD **104**, a display **105**, a keyboard **106**, a pointing device **107**, and a data communication unit **108**.

(26) The CPU (central processing unit/processor) **101** comprehensively controls the poster creation apparatus **100** and realizes operations according to the present embodiment, for example, by reading a program stored in the ROM **102** into the RAM **103** and executing it. Although only one CPU is shown in FIG. 1, there may be a plurality of CPUs.

(27) The ROM **102** is a general-purpose ROM and stores, for example, a program to be executed by the CPU **101**. The RAM **103** is a general-purpose RAM and is used, for example, as a working memory to temporarily store various information when a program is executed by the CPU **101**.

(28) The HDD (Hard Disk Drive) **104** is a storage medium (a storage unit) for storing image files, a database of results of image analysis and other processing, and skeletons used by the poster creation application.

(29) The display **105** is a display unit that displays a user interface (UI) according to the present embodiment and also displays image data (hereinafter, also referred to as an “image”) as a layout result of an electronic poster to the user. The keyboard **106** and the pointing device **107** accept instructional operations from the user. The display **105** may have a touch sensor function.

(30) The keyboard **106** is used, for example, by the user to enter the number of double-page poster sheets to be generated, on the UI shown on the display **105**.

(31) The pointing device **107** is used, for example, by the user to click a button on the UI displayed on the display **105**.

(32) The data communication unit **108** communicates with an external apparatus via a wired or wireless network. For example, the data communication unit **108** transmits data laid out by an automatic layout function to a printer or a server that can communicate with the poster creation apparatus **100**.

(33) The data bus **110** connects the blocks shown in FIG. 1 such that they can communicate with each other.

(34) Note that the configuration shown in FIG. 1 is only an example and is not limited to this. For example, the poster creation apparatus **100** may not have the display **105** and may display the UI

on an external display.

(35) In the present embodiment, the poster creation application is stored in the HDD **104**. The poster creation application is started when a user performs an operation such as clicking or double-clicking on an application icon displayed on the display **105** with the pointing device **107**.

(36) FIG. **2** is a software block diagram of a poster creation application. The poster creation application includes a poster creation condition specification unit **201**, a text specification unit **202**, image specification unit **203**, a target impression specification unit **204**, poster display unit **205**, and poster creation unit **210**. The poster creation unit **210** further includes an image acquisition unit **211**, an image analysis unit **212**, a skeleton acquisition unit **213**, a skeleton selection unit **214**, a color scheme pattern selection unit **215**, a font selection unit **216**, a layout unit **217**, an impression estimation unit **218**, and a poster selection unit **219**.

(37) When the poster creation application is installed in the poster creation apparatus **100**, a start icon is displayed on a top screen (desktop) of the OS (operating system) running on the poster creation apparatus **100**. When the user performs an operation (for example, double-clicking) on the start icon displayed on the display **105** with the pointing device **107**, the poster creation application program stored in the HDD **104** is loaded into the RAM **103** and executed by the CPU **101**. As a result, the poster creation application is started.

(38) Program modules corresponding to the respective components shown in FIG. **2** are included in the poster creation application described above. By executing these program modules, the CPU **101** realizes the functions of the components shown in FIG. **2**. In the following description of the respective components shown in FIG. **2**, it is assumed that the corresponding components execute various processes. FIG. **2** shows a software block diagram related, among various units, the poster creation unit **210** that provides the automatic poster creation function.

(39) The poster creation condition specification unit **201** specifies poster creation conditions to the poster creation unit **210** according to UI operations with the pointing device **107**. In the present embodiment, the poster size, the number of posters to be created, and the purpose category are specified as the poster creation conditions. The poster size may be specified by actual width and height values, or by a paper size such as A1, A2, or the like. The purpose category indicates what the poster is used for, such as a restaurant, a school event, sales, etc.

(40) The text specification unit **202** specifies text information to be placed on the poster by a UI operation using the keyboard **106**. The text information placed on the poster includes a character string representing, for example, the title, the date and time, the location, and the like. The text specification unit **202** outputs each piece of text information to the skeleton acquisition unit **213** and the layout unit **217** such that each piece of text information is associated with type information indicating a type of information such as title information, date and time information, location information, or the like.

(41) The image specification unit **203** specifies one or more pieces of image data to be placed on the poster from images stored in the HDD **104**. The specifying of the image data may be performed by specifying a device or a directory or other similar file system structure where one or more images are stored. Alternatively, accompanying information of images such as shooting date and time or attribute information may be specified. The image specification unit **203** outputs the file path of the specified image to the image acquisition unit **211**.

(42) The target impression specification unit **204** specifies the target impression of the poster to be created. The target impression is an impression that the created poster is finally required to hold. In the present embodiment, a strength of an impression is specified by performing a UI operation on a word expressing the impression using the pointing device **107**. Information indicating the target impression specified by the target impression specification unit **204** is shared by the skeleton selection unit **214**, the color scheme pattern selection unit **215**, the font selection unit **216**, and the poster selection unit **219**. The details of the impression will be described later.

(43) Next, the configuration of the poster creation unit **210** is described in detail.

(44) The image acquisition unit **211** acquires the one or more pieces of image data specified by the image specification unit **203** from the HDD **104**. The image acquisition unit **211** outputs the acquired image data to the image analysis unit **212**, and outputs the number of acquired images to the skeleton acquisition unit **213**. Examples of images stored in the HDD **104** are still images and frame images extracted from moving images. The still images and the frame images are obtained from an imaging device such as a digital camera, a smart device or the like. The imaging device may be included in the poster creation apparatus **100** or may be included in an external apparatus. Note that when the imaging device is an external apparatus, images are acquired via the data communication unit **108**. Alternatively, the still image may be an illustration image created by image editing software or a CG image created by CG generation software. The still image and the extracted image may be obtained from a network site or a server via the data communication unit **108** and a network. An example of the image obtained from a network site or a server is a social networking service image (hereinafter referred to as “SNS image”). The program executed by the CPU **101** analyzes the data attached to each image to determine the storage source. For example, SNS images may be acquired from SNS via the application, and acquisition sources may be managed in the application. The images are not limited to those described above, and other types of images may be obtained.

(45) The image analysis unit **212** analyzes the image data acquired from the image acquisition unit **211** by a method described later to acquire information indicating an image feature value described later. More specifically, the image analysis unit **212** executes an object recognition process, which will be described later, and acquires information indicating the image feature value of the image data. The image analysis unit **212** outputs the image data to the layout unit **217** such that the information indicating the acquired image feature value is associated with the image data.

(46) The skeleton acquisition unit **213** acquires from the HDD **104** one or more skeletons that meet the conditions specified by the poster creation condition specification unit **201**, the text specification unit **202**, and the image acquisition unit **211**. In the present embodiment, the skeleton is information representing the placement of character strings, images, graphics, etc., to be placed on the poster.

(47) FIGS. **3A** and **3B** are diagrams for illustrating an example of a skeleton. Three graphic objects **302** to **304**, one image object **305**, and four text objects **306** to **309** on which text information is placed are placed on the skeleton **301** shown in FIG. **3A**. In addition to the position, the size, and the angle of each object, metadata required to create a poster is also stored. FIG. **3B** is a diagram illustrating an example of metadata. For example, text objects **306** to **309** hold metadata of attributes indicating types of text information placed on the respective text objects. In this example, the metadata indicates that a title is placed on the text object **306**, a subtitle is placed on the text object **307**, and main texts are placed on the text objects **308** and **309**. The graphic objects **302** to **304** hold metadata of attributes indicating shapes of the graphics and color scheme numbers (color scheme IDs) indicating color scheme patterns. In this example, the metadata indicates that the graphic objects **302** and **303** are rectangles, and the graphic object **304** is an ellipse. It is assumed that the graphic object **302** is assigned the color scheme number 1, and the graphic objects **303** and **304** are assigned the color scheme number 2. The color scheme number is information to be referred to when a color scheme is assigned, which will be described later. Different colors are assigned to different color scheme numbers. Note that the types of objects and metadata are not limited to those described above. For example, there may be a map object for placement of a map, a barcode object for placement of a QR code (registered trademark) or a barcode. As for metadata of text objects, there may be metadata representing the space between lines and the space between characters. The purpose of the skeleton may be described in the metadata and may be used to control whether or not the skeleton is allowed to be used according to the purpose.

(48) The skeleton may be stored in, for example, a CSV format in the HDD **104**, or may be stored in a DB format such as SQL. The skeleton acquisition unit **213** outputs the one or more skeletons

acquired from the HDD **104** to the skeleton selection unit **214**.

(49) The skeleton selection unit **214** selects one or more skeletons that match the target impression specified by the target impression specification unit **204** from the skeletons acquired from the skeleton acquisition unit **213**, and outputs the selected skeletons to the layout unit **217**. Since the layout of the entire poster is determined by the skeleton, it is possible to increase the variations of the created posters by preparing various types of skeletons in advance.

(50) The color scheme pattern selection unit **215** acquires, from the HDD **104**, one or more color scheme patterns that match the target impression specified by the target impression specification unit **204**, and outputs the acquired color scheme patterns to the layout unit **217**. Note that the color scheme pattern refers to a combination of colors used in a poster.

(51) FIG. **4** is a diagram illustrating an example of a table of color scheme patterns. In the present embodiment, the color scheme pattern indicates a combination of four colors. In a color scheme ID column in FIG. **4**, IDs each uniquely identifying a color scheme pattern are described. In color #1 to color #4 columns, colors are specified by RGB color values of 0 to 255 wherein the respective RGB color values are described in parentheses in the order of R, G, B ((R, G, B)=(0 to 255, 0 to 255, 0 to 255)). In the present embodiment, each color scheme pattern is given by a combination of four colors, but the number of colors included in each color scheme pattern is not limited to four. Furthermore, the number of colors may vary depending on specific color scheme patterns.

(52) The font selection unit **216** selects and acquires, from the HDD **104**, one or more font patterns that match the target impression specified by the target impression specification unit **204**, and outputs the acquired font patterns to the layout unit **217**. The font pattern refers to a combination of at least one of a title font, a subtitle font, and a text font.

(53) The layout unit **217** generates one or more pieces of poster data by laying out various data on each skeleton of one or more skeletons acquired from the skeleton selection unit **214** such that the number of pieces of generated data is equal to or greater than the specified number of posters to be created. The layout unit **217** places, on each skeleton, the text acquired from the text specification unit **202** and the image data acquired from the image analysis unit **212**, and applies the color scheme pattern and the font respectively acquired from the color scheme pattern selection unit **215** and the font selection unit. The layout unit **217** outputs the generated one or more pieces of poster data to the impression estimation unit **218**.

(54) The impression estimation unit **218** estimates the impression of each piece of poster data of the plurality of pieces of poster data acquired from the layout unit **217**, and associates the estimated impression with each piece of poster data. The impression estimation unit **218** outputs one or more pieces of poster data associated with the estimated impression to the poster selection unit **219**.

(55) The poster selection unit **219** compares the estimated impression of each of the plurality of pieces of poster data associated with the estimated impression acquired from the impression estimation unit **218** with the target impression specified by the target impression specification unit **204**, and selects the poster data with which the estimated impression closest to the target impression is associated. The selection result is stored in the HDD **104**. The poster selection unit **219** outputs the selected poster data to the poster display unit **205**.

(56) The poster display unit **205** outputs a poster image to be displayed on the display **105** according to the poster data acquired from the poster selection unit **219**. For example, the poster image is represented in bitmap data. The poster display unit **205** displays the poster image on the display **105**.

(57) Although not shown in the figure, the poster creation application may have a function of editing the created poster after the creation result is displayed on the poster display unit **205** such that the layout, the colors, and the shapes, etc. of the images, the texts, and the graphics are edited according to a user operation so as to achieve a design desired by the user.

(58) When a function is provided that allows it to print poster data stored in the HDD **104** using a printer according to the conditions specified by the poster creation condition specification unit **201**,

the user can obtain the created poster in the printed form.

(59) Examples of Display Screens

(60) FIG. 5 is a diagram illustrating an example of an application start screen **501** provided by the poster creation application. The application start screen **501** is displayed on the display **105**. When the user sets poster creation conditions, texts, and images, which will be described later, via the application start screen **501**, the contents of the setting made by the user are acquired via the UI screen by the poster creation condition specification unit **201**, the image specification unit **203**, and the text specification unit **202**.

(61) A title box **502**, a subtitle box **503**, and a main text box **504** accept specifying of text information to be placed on the poster. Three types of text information are accepted in the present embodiment, but this is by way of example and not limitation. For example, additional text information indicating a location, date and time, or the like may be accepted. It is not necessary that all are specified, and some specification boxes may be blank.

(62) An image specification area **505** is an area in which images to be placed on the poster are displayed. An image **506** indicates a thumbnail of a specified image. An image addition button **507** is a button for adding an image to be placed on the poster. When the user presses the image addition button **507**, the image specification unit **203** displays a dialog screen for selecting a file stored in the HDD **104** and accepts a selection of an image file from the user. A thumbnail of the selected image is added to the image specification area **505**.

(63) Impression sliders (impression slider bars, or impression setting sliders) **508** to **511** are objects for setting the factors of the target impression of the poster to be created. For example, the impression slider **508** is used to set the target impression in terms of factor of luxury. As the slider is moved to the right, the target impression in terms of the factor of luxury of the poster increases, and as the slider is moved to the left, the target impression of the poster in terms of the factor of luxury decreases (becomes cheap). By combining the factors of the target impression set by the respective sliders, not only the factors of the target impression set by one slider, but also the factors of the target impression set by the other sliders are reflected in the target impression. For example, in a case where the user performs a user operation on the screen of the poster creation application to set the impression slider **508** to the right of the center of the slider and the impression slider **511** to the left of the center of the slider, a poster with an impression of high luxury and low stateliness, that is, with an elegant impression is created. In a case where the user performs a user operation to set the impression slider **508** to the right of the center of the slider and the impression slider **511** to the right of the center of the slider, a poster with an impression of high luxury and high stateliness, that is, with a gorgeous impression is created. As described above, by combining the target impression factors indicated by a plurality of impression sliders, even if a target impression factor such as “luxury” is set in common, it is possible to combine with it various target impression factors such as “elegant” and/or “gorgeous”. That is, the target impression determined by a plurality of factors indicating impressions, but may be determined by one factor indicating an impression. In the present embodiment, it is assumed that when the slider is set to the leftmost position, the impression has a value of -2 , and when it is set to the rightmost position, the impression has a value of $+2$, and the value is corrected to an integer value in the range from -2 to $+2$. The values correspond to impressions such that -2 corresponds to “low”, -1 to “rather low”, 0 to “neutral”, $+1$ to “rather high”, and $+2$ to “high”. It should be noted that the reason why the values are corrected to integer values in the range from -2 to $+2$ is to match the scale of the estimated impression described below thereby facilitating the distance calculation. The correction is not limited to the above manner. For example, the values may be normalized to values between 0 to 1 .

(64) Radio buttons **512** control enabling and disabling of the setting of respective corresponding target impressions. The user can set whether to enable or disable the setting of each target impression by pressing the corresponding radio button **512**. For example, when the radio button **512** is set to be disabled, the control of the corresponding impression is not performed. For

example, when a user wants to create a calm poster with low dynamism but does not care about any other impressions, he/she can turn off the radio buttons **512** except for the radio button for the dynamism to create a poster with an impression of low dynamism. In the example shown in FIG. 5, luxury and familiarity are set to be enabled, and dynamism and stateliness are set to be disabled. Thus, it is allowed to control all target impressions or a part of the target impressions to be used in the poster creation. That is, it is possible to flexibly control the target impressions in the poster creation. Note that when each slider is set to the leftmost position, the target impression is set to the same value as when the target impression is disabled (for example as in a case where when the slider **508** is set to the leftmost position, the impression of luxury is set to 0), the radio button **512** may be removed. In this case, the user can disable the setting of each target impression by setting the corresponding slider to the leftmost position.

(65) A size list box **513** is a list box for setting the size of the poster to be created. In response to a clicking operation by the user with the pointing device **107**, a list of available poster sizes is displayed and the user is allowed to select a desired size.

(66) A box **514** is for setting the number of posters to be created.

(67) A category list box **515** is used to set the purpose category of the poster to be created.

(68) A reset button **516** is a button for resetting the setting information on the application start screen **501**.

(69) When the user presses an OK button **517**, the poster creation condition specification unit **201**, the text specification unit **202**, the image specification unit **203**, and the target impression specification unit **204** output the settings made on the application start screen **501** to the poster creation unit **210**. As a result, the poster creation condition specification unit **201** acquires the size of the poster to be created specified in the size list box **513**, the number of posters to be created specified in the box **514** for specifying the number of posters to be created, and the purpose category of the poster to be created specified in the category list box **515**. The text specification unit **202** acquires text information to be placed on the poster from the title box **503**, the subtitle box **503**, and the main text box **504**. The image specification unit **203** acquires the image file path from which the image to be placed on the poster, specified in the image specification area **505**, is available. The target impression specification unit **204** acquires the target impressions of the poster to be created from impression sliders **508** to **511** and radio buttons **512**. Note that the poster creation condition specification unit **201**, the text specification unit **202**, the image specification unit **203**, and the target impression specification unit **204** may modify the values set on the application start screen **501**. For example, the text specification unit **202** may remove unnecessary leading or trailing blank characters from the input text information. The target impression specification unit **204** may correct the values of target impressions specified by the impression sliders **508** to **511**.

(70) FIG. 6 is a diagram illustrating an example of a poster preview screen displayed on the display **105** and a poster image created by the poster display unit **205** is displayed on the poster preview screen. When an OK button **309** on the application start screen **501** is pressed and the poster creating is completed, the screen displayed on the display **105** transitions to the poster preview screen **601**.

(71) A poster image **602** is a poster image output by the poster display unit **205**. The poster creation unit **210** creates posters such that the number of created posters is equal to or larger than the number of posters to be created specified by the poster creation condition specification unit **201**, and the created posters are displayed in a list as poster images **602** on the poster preview screen **601**. When the user clicks a poster with the pointing device **107**, the clicked poster is selected.

(72) When an edit button **603** is pressed, it becomes possible to edit the selected poster via a UI that provides an edit function (not shown).

(73) A print button **604** allows it to print the selected poster via a control UI of a printer (not shown).

(74) Quantification of Poster Impression

(75) A method for quantifying the impression of a poster is described below. This quantification process is necessary for the poster creation process and is performed as preprocessing of an impression estimation process in **S911** in FIG. 9A, which will be described later. The process of quantifying the impression of the poster is performed at the stage of developing the poster creation application by a vendor or the like that develops the poster creation application. The process of quantifying the impression of the poster may be executed by the poster creation apparatus **100** or may be executed by an information processing apparatus different from the poster creation apparatus **100**. Note that in a case where the process is performed by an information processing apparatus different from the poster creation apparatus **100**, the process is executed by a CPU of this information processing apparatus.

(76) The poster impression quantification process involves quantifying impressions that people have for various posters. At the same time, the correspondence between the poster image and the impression of the poster is derived. This makes it possible to estimate the impression of the poster from the generated poster image. If the impression can be estimated, it becomes possible to control the impression of the poster image by correcting the poster image, or to search for a poster image having a certain target impression. Note that the poster impression quantification process is executed, for example, by operating an impression learning application for learning the poster image impression in advance prior to the poster creation process in the poster creation apparatus.

(77) FIG. 7 is a flowchart illustrating a poster impression quantification process. The flowchart shown in FIG. 7 is realized, for example, by the CPU **101** loading a program stored in the HDD **104** into the RAM **103** and executing the program. The poster impression quantification process is performed according to the flowchart shown in FIG. 7, which is described below. Note that the symbol “S” in the description of each process indicates a step in the flowchart (the same applies hereinafter in this specification).

(78) In **S701**, the CPU **101** acquires a subjective evaluation of the impression of a poster.

(79) FIG. 8 is a diagram for explaining an example of a method of a subjective evaluation of an impression of a poster. The CPU **101** presents the poster to a human evaluator and obtains from the human evaluator a subjective evaluation of the impression of the poster. In the evaluation, a measurement method such as an SD (Semantic Differential) method or a Likert scale method can be used. FIG. 8 illustrates an example of a questionnaire using the SD method, in which adjective pairs expressing impressions are presented to a plurality of human evaluators, and adjective pairs evoked by the target poster are scored. After the subjective evaluation results for a plurality of posters are acquired from a plurality of human evaluators, the CPU **101** calculates the average value of the responses to each adjective pair, thereby obtaining a representative score value for the corresponding adjective pair. The subjective impression evaluation may be performed by a method other than the SD method, as long as words expressing impressions and the score values corresponding to them can be determined.

(80) In **S702**, the CPU **101** performs factor analysis on the subjective evaluation result acquired by the subjective evaluation acquisition unit. When subjective evaluation results are directly used, the number of dimensions is given by the number of adjective pairs, which results in complicated control. Therefore, it is desirable to reduce the number of dimensions to a small value using an analysis technique such as principal component analysis, factor analysis, or the like such that efficient analysis becomes possible. In the following description of the present embodiment, it is assumed that the dimensions are reduced such that the number of factors is reduced to four by factor analysis. Note that the number of factors varies depending on the selection of adjective pairs in the subjective evaluation and the method of factor analysis. It is also assumed that the output of factor analysis is normalized. That is, each factor is scaled to have a mean of 0 and a variance of 1 in the posters used for analysis. As a result, -2, -1, 0, +1, and +2 of the impressions specified by the target impression specification unit **204** can be directly corresponded to -26, -16, mean value,

+16, and +26 in each impression, which makes it easy to calculate the distance between the target impression and the estimated impression, as will be described in further detail later. In the present embodiment, the four factors are luxury, familiarity, dynamism, and stateliness shown in FIG. 5. The names of these factors are given for convenience to convey impressions to the user via the user interface, and each factor is composed of a plurality of adjective pairs that influence each other.

(81) In **S703**, the CPU **101** associated each poster image with the impression thereof. Although quantification can be performed for posters subjected to subjective evaluation by the above-described method, it is necessary to estimate the impression of posters that will be created without subjective evaluation. The correspondence between the poster image and the impression can be achieved by learning a model for estimating the impression from the poster image using, for example, a deep learning method using the convolution neural network (CNN) or a machine learning method using a decision tree, or the like. In the present embodiment, the impression learning unit performs supervised deep learning using the CNN with the poster image as input and the four factors as output. That is, subjectively evaluated poster images and corresponding impressions are learned as correct answers to create a deep learning model, and impressions are estimated by inputting an unknown poster image into the learning model.

(82) The deep learning model created above is stored in, for example, the HDD **104**, and the impression estimation unit **218** loads the deep learning model stored in the HDD **104** into the RAM **103** and executes it.

(83) The impression estimation unit **218** converts the poster data acquired from the layout unit **217** into an image of the poster and estimates the impression of the poster by causing the CPU **101** or the GPU **109** to operate the deep learning model loaded in the RAM **103**. Although the deep learning method is used in the present embodiment, the method is not limited to the deep learning. For example, in a case where a machine learning method such as a decision tree is used, feature values such as an average luminance value, an edge value, and/or the like are extracted from the poster image using image analysis, and a machine learning model may be created which estimates the impression based on these feature values.

(84) FIG. **9** is a flowchart illustrating a poster creation process performed by the poster creation unit **210** of the poster creation application. As described above, the flowchart shown in FIG. **9** is started when the user sets various setting items on the poster creation application and presses the OK button.

(85) The flowchart shown in FIG. **9** is realized, for example, by the CPU **101** loading a program stored in the HDD **104** into the RAM **103** and executing the program. In the present embodiment, it is assumed that each step of the flow is executed by components shown in FIG. **2** which are realized by the CPU **101** by executing the poster creation application. Referring to FIG. **9**, the poster creation process is described below. Note that the symbol “S” in the description of each process indicates a step in the flowchart (the same applies hereinafter in this specification).

(86) In **S901**, the poster creation application displays the application start screen **501** on the display **105**. The user inputs each setting using the keyboard **106** or the pointing device **107** via a UI screen of the application start screen **501**.

(87) In **S902**, the poster creation condition specification unit **201**, the text specification unit **202**, the image specification unit **203**, and the target impression specification unit **204** respectively acquire corresponding settings from the application start screen **501**.

(88) In **S903**, the skeleton selection unit **214**, the color scheme pattern selection unit **215**, and the font selection unit **216** respectively determine the number of skeletons, the number of color scheme patterns, and the number of fonts to be selected according to the number of posters to be created specified by the poster creation condition specification unit **201**. In the present embodiment, using a method described later, the layout unit **217** generates poster data for the specified number of skeletons×the specified number of color scheme patterns×the specified number of fonts. The number of skeletons, the number of color scheme patterns, and the number of fonts are set such that

the number of posters created here is greater than the number of posters to be created. In the present embodiment, the number of skeletons, the number of color scheme patterns, and the number of fonts are determined according to equation 1 shown below.

$$(89) \text{ Thenumberofselections} = \text{.Math.} \sqrt[3]{\text{Thenumberofcreations} \times 2} \text{.Math.} \quad (1)$$

(90) For example, when the number of posters to be created is 6, the number of selections is 3, the number of pieces of poster data to be generated by the layout unit **217** is 27, and the poster selection unit **219** selects 6 from them.

(91) Thus, the poster selection unit **219** can select posters whose overall impression better matches to the target impression from among the generated poster data whose number of pieces is equal to or greater than the number of posters to be created.

(92) In **S904**, the image acquisition unit **211** acquires the image data. More specifically, the image acquisition unit **211** reads the image file specified by the image specification unit **203** from the HDD **104** into the RAM **103**.

(93) In **S905**, the image analysis unit **212** executes an analysis process on the image data acquired in **S904** to acquire information indicating a feature value. Examples of information indicating a feature value include meta information stored in an image and information indicating an image feature value that can be obtained by analyzing the image. These pieces of information are used in an object recognition process performed as the analysis process.

(94) Note that in the present embodiment, the object recognition process is performed as the analysis process, but the analysis process is not limited to the object recognition process, and other analysis processes may be executed. Note that the process in **S905** may be omitted. Details of the process performed in **S905** by the image analysis unit **212** are described below.

(95) The image analysis unit **212** executes the object recognition process on the image acquired in **S904**. Here, a known method can be used for the object recognition process. In the present embodiment, an object is recognized by a discriminator generated by deep learning. The discriminator outputs a likelihood value of 0 to 1 as to whether a certain pixel constituting the image is a pixel constituting each object, and recognizes that an object exists in the image when the likelihood value exceeding a certain threshold value. By recognizing the object image, the image analysis unit **212** can acquire the types and positions of objects such as faces, pets such as dogs and cats, flowers, food, buildings, ornaments, and landmarks.

(96) In **S906**, the skeleton acquisition unit **213** acquires skeletons that meet various setting conditions. In the present embodiment, it is assumed that one skeleton is described in one file and stored in the HDD **104**. The skeleton acquisition unit **213** sequentially reads skeleton files from the HDD **104** into the RAM **103**, while keeping skeletons that meet the conditions in the RAM **103** and deleting skeletons that do not meet the conditions from the RAM **103**. FIG. 9B shows a flowchart of a condition judgement process performed by the skeleton acquisition unit **213**. The condition determination processing of the skeleton acquisition unit **213** is described below with reference to FIG. 9B.

(97) In **S921**, the skeleton acquisition unit **213** determines whether the size of each poster stored into the RAM **103** matches the poster size specified by the poster creation condition specification unit **201**. Although in this example it is checked whether the sizes match, it may be checked only whether aspect ratios match. In this case, the skeleton acquisition unit **213** acquires a skeleton whose size matches the poster size specified by the poster creation condition specification unit **201** when the coordinate system of the read skeleton is properly increased or reduced.

(98) In **S922**, the skeleton acquisition unit **213** determines whether the purpose category specified by the poster creation condition specification unit **201** matches the category of the skeleton. For skeletons that are used only for a specific purpose, the purpose categories thereof are described in the skeleton files such that the skeletons are not acquired unless the corresponding purpose category is selected. When a skeleton is specifically designed for a specific purpose, for example,

as in a case where a graphic image representing school is drawn on a skeleton or as in a case where a graphic image of sports equipment or the like is drawn on a skeleton, such a skeleton is allowed to be used only for the specific purpose and is prevented from being used for other categories. Note that in a case where the application category is not set on the application start screen **501**, **S922** is skipped.

(99) In **S923**, the skeleton acquisition unit **213** determines whether the number of read-in image objects of the skeleton is equal to the number of images acquired by the image acquisition unit **211**.

(100) In **S924**, the skeleton acquisition unit **213** determines whether the text object of the read-in skeleton matches the text information specified by the text specification unit **202**. More specifically, it is determined whether the type of the text information specified by the text specification unit **202** exists on the skeleton. For example, assume that text strings are specified in the title box **502** and the main text box **504** on the application start screen **501** and a blank is specified in the subtitle box **503**. In this case, all text objects in the skeleton are searched, and if a text object whose type is set as “title” in the metadata and a text object whose type is set as “text” in the metadata are both found, it is determined that the skeleton is suitable, but otherwise it is determined that the skeleton is unsuitable.

(101) As described above, the skeleton acquisition unit **213** keeps in the RAM **103** skeletons that match all set conditions in terms of the skeleton size, the purpose category, the number of image objects, and the type of the text object. Although the skeleton acquisition unit **213** checks all skeleton files stored in the HDD **104** in the present embodiment, this is only by way of example and not limitation. For example, the poster creation application may store in the HDD **104** a database that associates file paths of skeleton files with search conditions (the skeleton size, the number of image objects, and types of text objects). In this case, the skeleton acquisition unit **213** searches the database for skeleton files that match the conditions, and the skeleton acquisition unit **213** reads only skeleton files found in the search from the HDD **104** into the RAM **103**, which allows it to acquire the skeleton files at a high speed. Referring again to FIG. **9A**, the description is continued.

(102) In **S907**, the skeleton selection unit **214** selects, from the skeletons acquired in **S906**, skeletons that match the target impression specified by the target impression specification unit **204**. FIGS. **10A** to **10C** are diagrams illustrating a method of selecting skeletons by the skeleton selection unit **214**. FIG. **10A** is a diagram illustrating an example of a table that associates skeletons with impressions. In FIG. **10A**, skeleton file names are described in a skeleton name column. The luxury, familiarity, dynamism, and stateliness columns describe numerical values that indicate the degree to which the respective skeletons contribute to the respective impressions. The values correspond to impressions such that -2 corresponds to “low”, -1 to “rather low”, 0 to “neutral”, $+1$ to “rather high”, and $+2$ to “high”.

(103) First, the skeleton selection unit **214** calculates the distance between the target impression acquired from the target impression specification unit **204** and each impression described in a skeleton impression table shown in FIG. **10A**. For example, in a case where the target impression is specified as “luxury=+1, familiarity=-1, dynamism=-2, and stateliness=+2”, the distances are calculated by the skeleton selection unit **214** as shown in FIG. **10B**. Note that in the present embodiment, the distance is expressed in the Euclidean distance (note that hereinafter, the distance is expressed in the Euclidean distance unless otherwise specified). The smaller the distance, the closer the estimated impression is to the target impression. Next, the skeleton selection unit **214** selects the top N skeletons with the shortest distance values shown in FIG. **10B**. In the present embodiment, the skeleton selection unit **214** selects the top two skeletons. That is, skeleton selection unit **214** selects the skeleton #1 and the skeleton #4.

(104) Note that N may be set to a fixed value, or may be variable according to the condition specified by the poster creation condition specification unit **201**. For example, in a case where the number of posters to be created is specified as 6 in the box **514** for specifying the number of

posters to be created on the application start screen **501**, the poster creation unit **210** creates 6 posters. The layout unit **217**, which will be described later, generates a poster by combining a skeleton, a color scheme pattern, and a font respectively selected by the skeleton selection unit **214**, the color scheme pattern selection unit **215**, and the font selection unit **216**. For example, in a case where two skeletons, two color scheme patterns, and two fonts are selected, a total of as many posters as $2 \times 2 \times 2 = 8$ are created, which satisfies the condition that the number of posters to be created is 6. As described above, the number of skeletons N to be selected may be determined according to the conditions specified by the poster creation condition specification unit **201**.

(105) Note that the value range of each impression in the skeleton impression table in FIG. **10A** is not necessarily needed to be equal to the value range of the impression specified by the target impression specification unit **204**. In the present embodiment, the impression is specified by the target impression specification unit **204** in the value range from -2 to $+2$, but the value range in the skeleton impression table may be different from this value range. In this case, the distance described above is calculated after the value range of the skeleton impression table is scaled to match that of the target impression. The distance calculated by the skeleton selection unit **214** is not limited to the Euclidean distance, and any distance such as the Manhattan distance, the cosine similarity, or the like may be used as long as it is possible to calculate the distance between vectors. In a case where the radio button **517** is set to be OFF to disable a specific target impression, the calculation of the distance of that impression is not performed.

(106) The skeleton impression table can be generated by, for example, creating poster images based on respective skeletons for the fixed color scheme pattern, font, image and text data to be placed on the skeletons, and estimating the impressions of the poster images. The generated skeleton impression table is stored in the HDD **104**. That is, by estimating the impressions of poster images that use the same text colors, images, etc., but are different in the layout of the text and/or the images, it is possible to describe the characteristics relative to other skeletons in the table. In the describing of the characteristics in the table, it is desirable to normalize the impressions based on the overall impression. That is, it is desirable to perform a process to cancel the effects of the used color scheme patterns, images, etc. on the impressions by averaging the impressions of a plurality of poster images created using the plurality of color scheme patterns, images, etc. from the one skeleton. This makes it possible to tabulate the impact of the positions on the impression. For example, it is possible to describe that in the case of the skeleton with a small image, the impression is determined by graphic elements or text elements independent of images, or that the tilted placement of images or text provides strong dynamism. FIG. **10C** illustrates examples of skeletons respectively corresponding to skeletons #1 to #4 shown in FIG. **10A**. For example, in skeleton #1, image objects and text objects are regularly arranged, and the area of the image is small, and thus the dynamism is low. In skeleton #2, the graphic object and the image object are circular, and thus the familiarity is high but the stateliness is low. In skeleton #3, a large image object is placed over a large area and a tilted graphic object is superimposed on the image object, and thus a high dynamism is obtained. In skeleton #4, an image is placed over the entire skeleton, and a text object is placed in a minimized area, and thus a high stateliness is obtained but dynamism is low. As described above, in a case where a poster image includes characters or images, poster images with different target impressions are created depending on how the characters or images are arranged. Note that the method of generating the skeleton impression table is not limited to that described above. The impressions may be estimated from features of placement information such as areas and/or coordinates of images or title text. Furthermore, the impressions may be manually adjusted. The skeleton impression table is stored in the HDD **104**, and the skeleton selection unit **214** reads out the skeleton impression table from the HDD **104** into the RAM **103** and refers to it.

(107) In **S908**, the color scheme pattern selection unit **215** selects a color scheme pattern that matches the target impression specified by the target impression specification unit **204**. The color

scheme pattern selection unit **215** refers to the color scheme pattern impression table corresponding to the color scheme pattern, as in **S906**, and selects a color scheme pattern according to the target impression. FIG. **11A** illustrates an example of a color scheme pattern impression table that associates color scheme patterns with impressions. The color scheme pattern selection unit **215** calculates the distance between the target impression and the impressions described in the luxury to stateliness columns in FIG. **11A** for each color scheme pattern, and selects the top N color scheme patterns with the smallest distance. In the present embodiment, the top two color scheme patterns are selected. Like the skeleton impression table, the color scheme pattern impression table is obtained by creating posters with various color scheme patterns while fixing the skeleton, the font, and the image and estimating the impressions of the resultant posters and tabulating the impression tendencies of the color scheme patterns.

(108) In **S909**, the font selection unit **216** selects a combination of fonts that match the target impression specified by the target impression specification unit **204**. The font selection unit **216** refers to the impression table corresponding to the font, as in **S906**, and selects a font according to the target impression. FIG. **11B** illustrates an example of a font impression table that associates fonts with impressions. The font selection unit **216** calculates the distance between the target impression and the impressions described in the luxury to stateliness columns in FIG. **11B** for each font, and selects the top N fonts with the smallest distance. Like the skeleton impression table, the font impression table is obtained by creating posters using various fonts while fixing the skeleton, the color scheme pattern, and the image and estimating the impressions of the resultant posters and tabulating the impression tendencies of the fonts.

(109) In **S910**, the layout unit **217** sets text information, images, color schemes, and fonts for the skeleton selected by the skeleton selection unit **214**, and creates a poster.

(110) The process in **S910** and the process of the layout unit **217** are described in detail below with reference to FIG. **12**, FIG. **13**, FIGS. **14A** to **14C**, and FIGS. **15A** to **15C**. FIG. **12** is a software block diagram illustrating details of the layout unit **217**. The layout unit **217** includes a color scheme assignment unit **1201**, an image placement unit **1202**, an image correction unit **1203**, a font setting unit **1204**, a text placement unit **1205**, and a text decoration unit **1206**. FIG. **13** is a flowchart illustrating the details of the process in **S910**. FIGS. **14A** to **14C** are diagrams illustrating information input to the layout unit **217**. FIG. **14A** is a table summarizing text information specified by the text specification unit **202** and images specified by the image specification unit **203**. FIG. **14B** illustrates an example of a table indicating color scheme patterns acquired by the color scheme pattern selection unit **215**. FIG. **14C** illustrates examples of fonts acquired from the font selection unit **216**. FIGS. **15A** to **15C** are diagrams for illustrating processing steps performed by the layout unit **217**.

(111) First, referring to FIG. **13**, the process in **S910** is described in detail below.

(112) In **S1301**, the layout unit **217** lists all combinations of skeletons acquired from the skeleton selection unit **214**, color scheme patterns acquired from the color scheme pattern selection unit **215**, and fonts acquired from the font selection unit **216**. The layout unit **217** sequentially creates poster data for each combination in following layout processing. For example, in a case where the number of skeletons acquired from the skeleton selection unit **214** is 3, the number of color scheme patterns acquired from the color scheme pattern selection unit **215** is 2, and the number of fonts acquired from the font selection unit **216** is 2, the layout unit **217** generates as many pieces of poster data as $3 \times 2 \times 2 = 12$. Next, in **S1301**, the layout unit **217** selects one combination from the list of combinations, and executes the processes in **S1302** to **S1307**.

(113) In **S1302**, the color scheme assignment unit **1201** assigns color scheme patterns acquired from the color scheme pattern selection unit **215** to the skeletons acquired from the skeleton selection unit **214**.

(114) FIG. **15A** is a diagram illustrating an example of a skeleton. In the following description of the present embodiment, it is assumed by way of example that a color scheme pattern with a color

scheme ID of 1 shown in FIG. 14B is assigned to a skeleton 1501 shown in FIG. 15A. The skeleton 1501 in FIG. 15A includes two graphic objects 1502 and 1503, one image object 1504, and three text objects 1505 to 1507. First, the color scheme assignment unit 1201 assigns color schemes to the graphic objects 1502 and 1503. More specifically, based on a color scheme number described in metadata of each graphic object, a color of a color scheme pattern is assigned. For example, in a case where metadata of a text object indicates that the type is “title”, the color scheme assignment unit 1201 assigns, for example, the last color of the color scheme pattern to this text object. That is, in the present example, the color #4 is assigned to the text object 1505. Next, the text color for text placed on text objects other than those whose type is specified as “title” in the metadata is set based on the brightness of the background of the text objects. In the present embodiment, if the brightness of the background of the text object is lower than or equal to a threshold value, white is assigned to the text color, but otherwise black is assigned to the text color. FIG. 15B is a diagram showing the state of the skeleton 1508 after the color scheme assignment described above is performed. The color scheme assignment unit 1201 outputs the color-assigned skeleton data to the image placement unit 1202.

(115) In S1303, the image placement unit 1202 places the image data acquired from the image analysis unit 212 on the skeleton data acquired from the color scheme assignment unit 1201 based on the accompanying analysis information. In this example, the image placement unit 1202 assigns the image data 1401 to the image object 1504 on the skeleton. In a case where the image object 1504 and the image data 1401 have different aspect ratios, the image placement unit 1202 performs trimming such that the aspect ratio of the image data 1401 is equal to the aspect ratio of the image object 1504. More specifically, based on the object position obtained as a result of analyzing the image data 1401 by the image analysis unit 212, the trimming is performed so as to minimize the reduction in the object area caused by the trimming. Note that the trimming method is not limited to this, and other trimming methods may be used. For example, the center of the image may be trimmed, or a composition is devised such that the face position is in a triangular composition, or the like. The image placement unit 1202 outputs the resultant image-assigned skeleton data to the image correction unit 1203.

(116) In S1304, the image correction unit 1203 acquires the image-assigned skeleton data from the image placement unit 1202, and corrects the images placed on the skeleton. In the present embodiment, when the image resolution is insufficient, upsampling is performed using super-resolution processing. First, the image correction unit 1203 determines whether the images placed on the skeleton satisfy a predetermined resolution. For example, in a case where an image of 1600 pixels×1200 pixels is assigned to an area of 200 mm×150 mm on the skeleton, the print resolution of the image that will be obtained when the image is printed can be calculated according to equation 2.

$$(117) \frac{1600}{200 \div 25.4} \approx 203[\text{dpi}] \quad (2)$$

(118) Next, in a case where it is determined that the print resolution of the image is lower than a threshold value, the image correction unit 1203 increases the resolution by performing super-resolution processing. However, in a case where it is determined that the print resolution of the image is greater than or equal to the threshold value and thus the resolution is sufficient, no particular image correction is performed. In the present embodiment, the super-resolution processing is performed when the print resolution is lower than 300 dpi.

(119) In S1305, the font setting unit 1204 sets the font acquired from the font selection unit 216 to the image-corrected skeleton data acquired from the image correction unit 1203. FIG. 14C shows an example of a combination of fonts selected by the font selection unit 216. In the following description of the present example, it is assumed that the font with a font ID of 2 shown in FIG. 14C is assigned to the image-corrected skeleton data. In this example, fonts are set for the text objects 1505, and 1506, and 1507 of the skeleton 1508. As for fonts of posters, in many cases, a

conspicuous font is used for titles to provide high visual attractiveness, while an easy-to-read font is used for other types of text to provide high visibility. Therefore, in the present embodiment, the font selection unit **216** selects two types of fonts, that is, a title font and a text font. The font setting unit **1204** sets the title font for the text object **1505** whose type is title, and sets the text font for the other text objects **1506** and **1507**. The font setting unit **1204** outputs the font-set skeleton data to a text placement unit **1205**. In the present embodiment, the font selection unit **216** selects two types of fonts, but this is only by way of example and not limitation. For example, only the title font may be selected. In this case, the font setting unit **1204** uses a font corresponding to the title font as the text font. That is, for example, in a case where a Gothic family font is used for the title, a Gothic family font that is highly readable may be used for other types of text objects, while in a case where a Mincho family font is used for the title, a Mincho family font may be used for other types of text objects, and so on. Of course, the title font and the text font may be the same. Alternatively, different fonts may be used depending on how prominent the text objects are to be. For example, the title font is used for the title text object and subtitle text objects, the text font is used for the other text objects, the title font is used for text objects described in fonts with a size greater than or equal to a predetermined value.

(120) In **S1306**, the text placement unit **1205** places the text specified by the text specification unit **202** on the font-set skeleton data acquired from the font setting unit **1204**. In the present embodiment, each text shown in FIG. **14A** is assigned according to the attribute described in the metadata of each text object of the skeleton. More specifically, “GREAT SUMMER SALE” whose attribute is title is assigned to the text object **1505**, and “BLOW OFF THE MIDSUMMER HEAT” whose attribute is subtitle is assigned to the text object **1506**. Nothing is set to the main text, and thus nothing is assigned to the text object **1507**. FIG. **15C** shows a skeleton **1509**, which is an example of skeleton data processed by the text placement unit **1205**. The text placement unit **1205** outputs the skeleton data to which the text has been placed to a text decoration unit **1206**.

(121) In **S1307**, the text decoration unit **1206** decorates the text object in the text-placed skeleton acquired from the text placement unit **1205**. In the present embodiment, if the color difference between the characters of the title and its background area is smaller than or equal to a threshold value, a process is performed to border the characters of the title. This improves the readability of the title. The text decoration unit **1206** outputs the decorated skeleton data, that is, the poster data that has been completely laid out to the impression estimation unit **218**.

(122) In **S1308**, the layout unit **217** determines whether all poster data has been generated. In a case where the layout unit **217** determines that poster data has been generated for all combinations of skeletons, color scheme patterns, and fonts, the layout unit **217** ends the layout processing and advances the processing flow to **S911**. In a case where it is determined that the poster data has not been generated for all combinations, the processing flow returns to **S1301** to generate poster data for a combination that has not yet been subjected to the generation.

(123) The process in **S910** has been described above. Referring again to FIG. **9A**, the description is continued.

(124) In **S911**, the impression estimation unit **218** renders each piece of poster data acquired from the layout unit **217**, and associates the estimated impression obtained by estimating the impression of the rendered poster image with the poster data. Note that the rendering process is a process of converting poster data into image data. For example, even if the color scheme pattern is the same, the layout can change depending on the skeleton, and thus which color is actually used in how large an area is different depending on the skeleton. Therefore, it is necessary to evaluate the final overall impression of the poster as well as the individual impressions of the color scheme patterns and skeletons. Therefore, this process is performed at this timing.

(125) This makes it possible to evaluate not only the impression of individual elements of the poster such as color schemes and positions, but also the impression of the final poster including laid images and characters.

(126) In **S912**, the poster selection unit **219** selects a poster to be output to the display **105** (thereby being presented to the user) based on the poster data and the associated estimated impression acquired from the impression estimation unit **218**. In the present embodiment, the poster selection unit **219** selects a poster in which the value of the distance between the target impression and the estimated impression of the poster is smaller than or equal to a predetermined threshold value.

(127) Note that in the present embodiment, the distance is expressed in the Euclidean distance. The smaller the Euclidean distance, the closer the estimated impression is to the target impression. The distance calculated by the poster selection unit **219** is not limited to the Euclidean distance, and any distance such as the Manhattan distance, the cosine similarity, or the like may be used as long as it is possible to calculate the distance between vectors.

(128) In a case where the number of selected posters is less than the number of posters to be created specified by the poster creation condition specification unit **201**, the poster selection unit **219** further selects a missing number of posters in ascending order of the value of the distance between the target impression and the estimated impression of the poster. In the present embodiment, the poster selection unit **219** also selects the missing posters, but this is by way of example and not limitation. For example, in a case where the number of posters selected by the poster selection unit **219** is less than the number of posters to be created, the preview screen **601** may display that the number of posters is insufficient. Alternatively, the poster selection unit **219** may select the missing posters and display them on the preview screen **601** such that the posters whose value of the distance between the target impression and the estimated impression is less than or equal to the threshold value can be distinguished from the posters whose value of the distance is greater than the threshold value. Still alternatively, in the case where the number of selected posters is insufficient, the process may return to **S903** and increase the number of selected skeletons, color scheme patterns, and fonts.

(129) In **S913**, the poster display unit **205** renders the poster data selected by the poster selection unit **219** and outputs the resultant poster image to the display **105**. That is, the preview screen **601** in FIG. 6 is displayed.

(130) Above is described the process of creating the posters according to the impression specified by the user.

(131) As described above, according to the present embodiment, it is possible to create a poster that expresses the impression desired by the user. More specifically, in the present embodiment, a plurality of variations of poster candidates can be created according to the target impression by combining elements that make up the posters, such as skeletons, color scheme patterns, and fonts, based on the target impression. Furthermore, by estimating the overall impression of the poster and selecting a poster with an impression close to the target impression from the one or more candidate posters, it is possible to create a poster that meets the user's intentions in terms of not only individual elements, but also the overall impression. More specifically, for example, assume that the target impression is specified on the application start screen **501** such that -1 is set to luxury, $+1$ is set to familiarity, and dynamism and stateliness are disabled. In this case, for example, the poster image **602** on the preview screen **601** is generated such that the luxury value is -1.2 , the familiarity value is $+0.9$, the dynamism value is $+0.2$, and the stateliness value is -1.3 , and thus the estimated impression is close to the target impression.

First Modification of First Embodiment

(132) In the first embodiment described above, the target impression is set by using the impression sliders **508** to **511** on the application start screen **501** as objects for setting the target impression, but the method for setting the target impression is not limited to this.

(133) Referring to FIGS. **16A** to **16D**, examples of UIs for setting a target impression are described. FIG. **16A** illustrates an example of setting a target impression with a UI on a radar chart. By operating a handle **1601** on the radar chart in FIG. **16A**, the target impression can be set along each axis. The target impression specification unit **204** acquires a target impression, for example, such

that -2 is obtained when the handle **1601** is at the center of the UI, and +2 is obtained when it is at an outermost position. In the example shown in FIG. **16A**, the target impressions are specified such that luxury=+0.8, familiarity=+1.1, dynamism=-0.1, and stateliness=-0.7. As in this example, the target impression may be expressed as a decimal fraction. FIG. **16B** shows an example of a radar chart in which some target impressions are disabled. For example, if the user double-clicks the handle **1601** with the pointing device **107**, the target impression of the axis on which the handle **1601** is located is disabled and hidden. When the user clicks again the axis **1602** of the radar chart with the pointing device **107**, the target impression is enabled and is displayed. In the example in FIG. **16B**, although dynamism is disabled, the target impression is the same as that in FIG. **16A**, except for dynamism.

(134) FIG. **16C** shows an example of a UI for setting a target impression based on images instead of words. In a sample poster display area **1603**, poster images **1604** to **1607** are arranged that differ greatly from each other with respect to an impression item. A check box **1608** is displayed for each poster image. If the user turns on a check box **1608** by clicking a poster that is close to a poster the user wants to create by using the pointing device **107**, the poster is selected. The target impression specification unit **204** determines the target impression by referring to the impression corresponding to the selected poster image.

(135) FIG. **16D** illustrates a table representing impressions corresponding to the poster images **1604** to **1607** shown in FIG. **16C** and final target impressions. The luxury, familiarity, dynamism, and stateliness columns describe numerical values that indicate the degree to which the poster image contributes to each impression. For example, assume that poster images **1604** and **1607** are selected as shown in FIG. **16C**. In this case, the target impression specification unit **204** determines, as the impression, an impression obtained as a result of synthesizing the impressions of the poster images **1604** and **1607**. In this example, among the factors corresponding to the selected poster images, the value having the maximum absolute value is taken as the value of each factor of the target impression. Although the example described above presents a poster image with a maximum impression value, this is by way of example and not limitation. For example, a poster image may be used in which a plurality of impressions have large values, or poster images may be presented such the number of presented poster images is equal to or greater than the number of impressions. This allows the user to intuitively specify the target impression based on the actual posters instead of using words.

Second Embodiment

(136) In the first embodiment described above, constituent elements of a poster such as a skeleton, a color scheme pattern, and a font are selected based on an image and a target impression, and the poster is created using the selected elements. In a second embodiment described below, a combination generation unit searches, according to a genetic algorithm, for a combination of constituent elements of a poster that gives an overall impression of the poster close to a target impression. This makes it possible to more flexibly select optimal poster constituent elements for the target impression without pre-calculating a skeleton impression table, a color scheme pattern impression table, or a font impression table.

(137) FIG. **17** is a software block diagram of a poster creation application according to a second embodiment. In the configuration of the block diagram shown in FIG. **17**, a combination generation unit **1701** is used instead of the skeleton selection unit **214**, the color scheme pattern selection unit **215**, and the font selection unit **216** shown in FIG. **2**. The constituent elements assigned the same reference numbers as those in FIG. **2** perform the same processing as the processing according to the first embodiment described above, and therefore a duplicate description thereof is omitted.

(138) The combination generation unit **1701** acquires one or more skeletons from the skeleton acquisition unit **213**, poster data and estimated poster impression from the impression estimation unit **218**, a target impression from the target impression specification unit **204**. The combination generation unit **1701** also acquires a list of color scheme patterns and fonts from the HDD **104**. The

combination generation unit **1701** generates a combination of constituent elements of a poster (skeletons, color scheme patterns, fonts) used for poster creation. The combination generation unit **1701** outputs the generated combination of poster constituent elements to the layout unit **217**.

(139) The poster selection unit **1702** selects a poster whose distance between the estimated impression of the poster and the target impression specified by the target impression specification unit **204** is smaller than or equal to a threshold value from the poster data acquired from the impression estimation unit **218**, and stores the selected poster in the RAM **103**. The poster selection unit **1702** determines whether the number of selected and stored posters reaches the number of posters to be created specified in the box **514** for specifying the number of posters to be created.

(140) FIG. **18** is a flowchart illustrating a process performed by the poster creation unit **210** of the poster creation application according to the present embodiment. Among the processing steps in this flowchart, the processing steps assigned the same step numbers as those in the flowchart shown in FIG. **9** are similar to the processing steps described in the first embodiment, and thus a further description thereof is omitted. Note that in the process according to this flowchart, **S903** and **S907** to **S909** shown in FIG. **9** are omitted.

(141) The process in **S1801** is described below for two cases: the process is executed for the first time; and the process is executed in a second and subsequent executions in an iterative calculation loop. First, when **S1801** is executed for the first time, the combination generation unit **1701** acquires tables of skeletons, color schemes, and fonts used for poster creation. FIGS. **19A** to **19D** show tables used by the combination generation unit **1701**. FIG. **19A** shows a list of skeletons acquired by the combination generation unit **1701** from the skeleton acquisition unit **213**. FIGS. **19B** and **19C** show, respectively, a list of fonts and a list of color schemes acquired by the combination generation unit **1701** from the HDD **104**. The combination generation unit **1701** generates random combinations from the three tables described above. In the present embodiment, 100 combinations are generated. FIG. **19D** shows a combination table generated according to the present embodiment.

(142) After that, the combination generation unit **1701** performs the processes in **S910**, **S911**, and **S1802** for all the generated combinations.

(143) In **S1801** in second and subsequent executions in the iterative calculation loop, the combination generation unit **1701** calculates the value of the distance between the target impression and the estimated poster impression acquired from the impression estimation unit **218**, and associates the calculated distance with the combination table. FIG. **20** is a diagram for explaining the process in **S1801** performed in second and subsequent executions in the iteration. FIG. **20A** shows a table obtained as a result of associating the value of the distance between the target impression and the estimated impression of the poster to the table shown in FIG. **19D**. More specifically, the layout unit **217** creates a poster based on the combination table shown in FIG. **19D**, and the impression estimation unit **218** estimates the impression of each created poster. The distance column in FIG. **20A** describes the value of the distance between the target impression and the estimated impression of the poster created for the combination indicated in the corresponding row. The combination generation unit **1701** generates a new combination table from FIG. **20A**. FIG. **20B** shows a newly generated combination table. In the present embodiment, the new combination is generated by tournament selection and uniform crossover in a genetic algorithm. In this process, first, N combinations are randomly selected from the table shown in FIG. **20A**. For example, N=3. Next, from among the selected combinations, the top two combinations with the shortest distance (that provides impressions close to the target impression) are selected. Finally, a new combination is generated by randomly replacing elements of the combinations (skeleton IDs, color scheme IDs, font IDs) across the two selected combinations. For example, combination IDs **1** and **2** in the table shown FIG. **20B** result from the combination IDs **1** and **3** in FIG. **20A** with the color scheme IDs interchanged. The total table shown in FIG. **20B** is obtained as a result of generating new 100 combinations by repeatedly performing the procedure described above.

(144) Thus, it becomes possible to efficiently search for a combination based on the value of the distance between the target impression and the estimated impression. Although 100 combinations are generated in the present embodiment, the number of combinations is not limited to 100. Furthermore, although the tournament selection and the uniform crossover are used, other methods such as ranking selection, roulette selection, and one-point crossover may be used. In addition, a mutation may be incorporated into the process to make it difficult to fall into a local optimum solution. Although skeletons (arrangements), color scheme patterns, and fonts are used as constituent elements of a poster to be searched, other constituent elements may be used. For example, a plurality of patterns may be prepared to be inserted into the background of a poster, and a determination as to which pattern to use and which not to use may be made by searching. By increasing the number of constituent elements to be searched, it becomes possible to create a greater variety of posters and increase the range of impressions of the posters.

(145) In **S1802**, the poster selection unit **1702** calculates the value of the distance between the estimated poster impression and the target impression in the same manner as in **S1801**, and generates a table similar to that shown in FIG. 20A. The poster selection unit **1702** stores, in the RAM **103**, the poster images whose value of the distance to the target impression is smaller than or equal to a threshold value.

(146) In **S1803**, the poster selection unit **1702** determines whether the number of poster images stored in the RAM **103** in **S1802** has reached the number of posters to be created specified in the box **514** for specifying the number of posters to be created. In a case where the poster selection unit **1702** determines that the number of poster images has reached the number of posters to be created, the poster selection unit **1702** ends the poster creation process, but otherwise the poster selection unit **1702** returns to **S1801**. That is, the second iteration processing from **S1801** to **S1802** is performed. The iteration processing is performed repeatedly until as many poster images with a distance to the target impression smaller than or equal to the threshold value as the specified number of posters to be created have been generated and stored in the RAM **103**. When poster images with a distance to the target impression smaller than or equal to the threshold value are generated and stored and the number of poster images reaches or exceeds the specified number of posters to be created, the poster selection unit **1702** may compare the distance values of the stored poster images with each other and may finally keep only poster images with smaller distances in the RAM **103**. In this case, poster images that are determined to have larger distance values as a result of the comparison may be deleted from the RAM **103**.

(147) In the present embodiment, the genetic algorithm is used to search for combinations of poster constituent elements, but the search method is not limited to this, and other search methods such as a neighborhood search method, a tab search method, or the like may be used.

(148) As described above, according to the present embodiment, by searching for a combination of constituent elements to be used in a poster, it is possible to create a poster with an overall impression close to the target impression. This is particularly effective when a poster is created in accordance with an image and text information input by a user. For example, consider a case where it is desired to create a poster such that an image has a dynamic impression but the poster as a whole has a calm impression. In the present embodiment, it is possible to evaluate the overall impression of the poster and search for a combination of a skeleton, a color scheme pattern, and a font that provides an impression close to a target impression. Therefore, the elements of the poster may be controlled depending on the image. For example, to reduce the impact of a particular image, a skeleton with a small image area and/or a more subdued font or color scheme may be used. According to the present embodiment, it is possible to flexibly find a combination of constituent elements that are optimal for the overall impression of the poster, and it is possible to create posters various variations that are close to the target impression.

Third Embodiment

(149) In the examples according to the first and second embodiments, a poster is created by

controlling constituent elements of the poster based on the target impression. A third embodiment discloses an example in which templates obtained by combining skeletons, color scheme patterns, and fonts are prepared in advance, and a poster is created only by the layout unit setting an image and text information. This makes it possible to create a poster that matches the target impression by performing simpler processing.

(150) FIG. 21 is a software block diagram of a poster creation application according to a third embodiment. In the configuration of the block diagram shown in FIG. 21, a template acquisition unit 2101 is used instead of the skeleton acquisition unit 213, the skeleton selection unit 214, the color scheme pattern selection unit 215, and the font selection unit 216 shown in FIG. 2. The constituent elements assigned the same reference numbers as those in FIG. 2 perform the same processing as the processing according to the first embodiment described above, and therefore a duplicate description thereof is omitted.

(151) A template acquisition unit 2101 acquires from the HDD 104 one or more templates that meet the conditions specified by poster creation condition specification unit 201, the text specification unit 202, and the image acquisition unit 211. In the present embodiment, the template refers to a skeleton for which a color scheme and a font have been set in advance. The template acquisition unit 2101 outputs the acquired one or more templates to the layout unit 2102.

(152) The layout unit 2102 creates poster data by laying out the image obtained from the image acquisition unit 211 and the text obtained from the text specification unit 202 on each template acquired from the template acquisition unit 2101. The layout unit 2102 outputs the created poster data to the impression estimation unit 218.

(153) FIG. 22 is a flowchart illustrating a process performed by the poster creation unit 210 of the poster creation application according to the present embodiment. The constituent elements assigned the same reference numbers as those in FIG. 9 perform the same processing as the processing according to the first embodiment described above, and therefore a further duplicate description thereof is omitted. In the process according to this flowchart, S906 to S909 shown in FIG. 9 are omitted.

(154) In S2201, a template acquisition unit 2101 acquires from the HDD 104 one or more templates that meet the conditions specified by poster creation condition specification unit 201, the text specification unit 202, and the image acquisition unit 211. In the present embodiment, the template refers to a skeleton for which a color scheme and a font have been set in advance. It is assumed that each template is described in one file and stored in the HDD 104.

(155) The template acquisition unit 2101, like the skeleton acquisition unit 213, sequentially reads template files from the HDD 104 into the RAM 103, while keeping templates that meet the setting conditions in the RAM 103 and deleting templates that do not meet the conditions from the RAM 103. The template acquisition unit 2101 outputs the acquired one or more templates to the layout unit 2102.

(156) In S2202, the layout unit 2102 creates poster data by laying out the image obtained from the image analysis unit 211 and the text obtained from the text specification unit 202 on each template acquired from the template acquisition unit 2101. The layout unit 2102 outputs the created one or more pieces of poster data to the impression estimation unit 218. The setting of the image is performed in a similar manner to that by the image placement unit 1202, and the setting of the text information is similar to that by the text placement unit 1205, and thus duplicate descriptions thereof are omitted.

(157) As described above, according to the present embodiment, by preparing templates in advance on which various color schemes and fonts are set, it is possible to create a poster that provides an impression close to the target impression only by performing simple processing.

OTHER EMBODIMENTS

(158) Embodiment(s) of the present disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more

programs) recorded on a storage medium (which may also be referred to more fully as a ‘non-transitory computer-readable storage medium’) to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like. While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the present disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(159) This application claims the benefit of Japanese Patent Application No. 2022-104050 filed Jun. 28, 2022, which is hereby incorporated by reference herein in its entirety.

Claims

1. An information processing apparatus, comprising: at least one processor; and a memory that stores a program which, when executed by the at least one processor, causes the at least one processor to function as: an acceptance unit configured to accept from a user a target impression of a poster image to be created; and a creation unit configured to create the poster image based on the target impression, wherein information indicating a difference between an impression that the poster image created by the creation unit has and the specified target impression is less than a predetermined threshold value.
2. The information processing apparatus according to claim 1, further comprising a first display unit configured to display a screen for accepting the target impression from the user, wherein the acceptance unit accepts specifying the target impression displayed on the screen from the user.
3. The information processing apparatus according to claim 2, wherein the first display unit displays a slider on the screen as an object for specifying the target impression, and the acceptance unit accepts the target impression specified by operating the slider from the user.
4. The information processing apparatus according to claim 2, wherein the first display unit displays a radar chart on the screen as an object for specifying the target impression, and the acceptance unit accepts the target impression specified by operating the radar chart from the user.
5. The information processing apparatus according to claim 1, wherein the target impression is determined by a combination of factors indicating the impression.
6. The information processing apparatus according to claim 1, wherein the poster image is created by changing a color scheme of the poster image to match a wide variety of the target impressions.
7. The information processing apparatus according to claim 1, wherein in a case where the poster image includes a character, the poster image is created by changing a font of the character to match a wide variety of the target impressions.
8. The information processing apparatus according to claim 1, wherein in a case where the poster image includes a character or an image, the poster image is created by changing a manner of

arranging the character or the image to match a wide variety of the target impressions.

9. The information processing apparatus according to claim 1, wherein the poster image is created based on a skeleton which is information indicating arrangements of a character, an image, and a graphic on the poster image, and the poster image is created by changing the skeleton to match a wide variety of the target impressions.

10. The information processing apparatus according to claim 9, wherein the poster image is created by arranging a character and an image on a template that is prepared in advance as a combination of the skeleton with a color scheme and a font of the character in the poster image.

11. The information processing apparatus according to claim 9, wherein the poster image is created by selecting the skeleton that provides an impression close-similar to the target impression, and a color scheme and a font in the poster image, and based on a combination of the selected skeleton and the color scheme and the character font in the poster image.

12. The information processing apparatus according to claim 1, wherein the at least one processor further functions as a selection unit configured to select, in a case where a plurality of the poster images are created, as many or more poster images having an impression similar to the target impression from the plurality of the poster images as the number of posters to be created specified by a user, and a second display unit configured to display the poster images, wherein the second display unit displays as many or more poster images having an impression close to the target impression as the number of posters to be created specified by a user.

13. The information processing apparatus according to claim 12, wherein the selection unit selects a poster image created by the creation unit that corresponds to a value less than a predetermined threshold value in terms of information indicating a difference between an impression that the poster image has and the specified target impression.

14. The information processing apparatus according to claim 12, wherein the selection units selects poster images created by the creation unit that correspond to values less than a predetermined threshold value in terms of information indicating a difference between an impression that the poster image has and the specified target impression in ascending order of the information indicating the difference from the target impression.

15. A control method for an information processing apparatus, comprising: accepting from a user a target impression of a poster image to be created, wherein the target impression is received from the user via a user interface (UI) entered through a display; and creating the poster image based on the target impression, wherein information indicating a difference between an impression that the poster image created by the creation unit has and the specified target impression is less than a predetermined threshold value.

16. A non-transitory computer-readable storage medium storing a program for causing a computer to execute a control method for an information processing apparatus comprising: accepting from a user a target impression of a poster image to be created; and creating the poster image based on the target impression, wherein information indicating a difference between an impression that the poster image created by the creation unit has and the specified target impression is less than a predetermined threshold value.
